Tutorial for Chapter 4

Exercise 1.
Let X be a discrete random variable with the following probability distribution

x	0	1	2	3
$p(x)$	$\frac{1}{125}$	$\frac{12}{125}$	$\frac{48}{125}$	$\frac{64}{125}$

Find $E(X)$ and $E(X^2)$

Solution 1.

- The mean

$$\begin{aligned} E[X] &= \sum_{x=0}^{3} xp(x) = 0 \cdot \frac{1}{125} + 1 \cdot \frac{12}{125} + 2 \cdot \frac{48}{125} + 3 \cdot \frac{64}{125} \\ &= \frac{300}{125} = \frac{12}{5} = 2.4 \end{aligned}$$

- The second Moment

$$\begin{aligned} E\left[X^2\right] &= \sum_{x=0}^{3} x^2 p(x) = 0^2 \cdot \frac{1}{125} + 1^2 \cdot \frac{12}{125} + 2^2 \cdot \frac{48}{125} + 3^2 \cdot \frac{64}{125} \\ &= \frac{780}{125} = 6.24 \end{aligned}$$

Exercise 2.
An insurance policy pays 100 per day for up to three days of hospitalization and 50 per day for each day of hospitalization thereafter.

The number of days of hospitalization, X, is a discrete random variable with probability function

$$P(X = k) = \begin{cases} \frac{6-k}{15} & k = 1, 2, 3, 4, 5 \\ 0 & \text{elsewhere} \end{cases}$$

Determine the expected payment for hospitalization under this policy.

Solution 2.
The probabilities of $1, 2, 3, 4,$ and 5 days of hospitalization are $5/15, 4/15, 3/15, 2/15,$ and $1/15$ respectively.

The payments are $100, 200, 300, 350,$ and 400 respectively.

The expected value is

$$\frac{100(5) + 200(4) + 300(3) + 350(2) + 400(1)}{15} = 220.$$

Exercise 3.
Let X have the following probability density

$$f(x) = \begin{cases} \frac{x}{2} & \text{for } 0 < x \leq 1 \\ \frac{1}{2} & \text{for } 1 < x \leq 2 \\ \frac{3-x}{2} & \text{for } 2 < x < 3 \\ 0 & \text{elsewhere} \end{cases}$$

Find the expected value of $g(X) = X^2 - 5X + 3$.

Solution 3.

- The mean

$$E[X] = \int_0^1 \frac{x^2}{2} dx + \int_1^2 \frac{x}{2} dx + \int_2^3 \frac{3x - x^2}{2} dx = \frac{3}{2}$$

- The second moment

$$E\left[X^2\right] = \int_0^1 \frac{x^3}{2} dx + \int_1^2 \frac{x^2}{2} dx + \int_2^3 \frac{3x^2 - x^3}{2} dx = \frac{8}{3}$$

- Finally,

$$\begin{aligned} E\left[g(X)\right] &= E\left(X^2 - 5X + 3\right) \\ &= E\left[X^2\right] - 5E\left[X\right] + 3 \\ &= \frac{8}{3} - 5 \cdot \frac{3}{2} + 3 = -\frac{11}{6} \end{aligned}$$

Exercise 4.
Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{|x|}{10} & -2 \leq x \leq 4 \\ 0 & \text{elsewhere} \end{cases}$$

Calculate the expected value of X.

Solution 4.
The formula for the expected value of a continuous random variable is:

$$E[X] = \int_{-\infty}^{\infty} x f(x) \, dx$$

Given the density function $f(x)$, we have:

$$E[X] = \int_{-2}^{4} x f(x) \, dx = \int_{-2}^{4} x \left(\frac{|x|}{10}\right) \, dx$$

Since $f(x)$ contains the absolute value, we break the integral into two parts:

- For $-2 \leq x < 0$, $|x| = -x$.
- For $0 \leq x \leq 4$, $|x| = x$.

Thus, we can express $E[X]$ as:

$$
\begin{aligned}
E[X] &= \int_{-2}^{0} x\left(\frac{-x}{10}\right)\,dx + \int_{0}^{4} x\left(\frac{x}{10}\right)\,dx \\
&= -\frac{1}{10}\int_{-2}^{0} x^2\,dx + \frac{1}{10}\int_{0}^{4} x^2\,dx \\
&= -\frac{1}{10}\left[\frac{x^3}{3}\right]_{-2}^{0} + \frac{1}{10}\left[\frac{x^3}{3}\right]_{0}^{4} \\
&= -\frac{4}{15} + \frac{32}{15} \\
&= \frac{28}{15}
\end{aligned}
$$

Exercise 5.
Find μ'_r and σ^2 for the random variable X that has the probability density

$$
f(x) = \begin{cases} \frac{1}{\ln 3} \cdot \frac{1}{x} & \text{for } 1 < x < 3 \\ 0 & \text{elsewhere} \end{cases}
$$

Solution 5.

- The rth moment about the origin

$$
\mu'_r = \frac{1}{\ln 3}\int_1^3 x^{r-1}dx = \frac{1}{\ln 3}\frac{x^r}{r}\bigg|_1^3 = \frac{1}{r\ln 3}\cdot(3^r - 1) = \frac{3^r - 1}{r\ln 3}.
$$

- The mean and second moment are

$$
\mu = \frac{2}{\ln 3}, \quad \mu'_2 = \frac{8}{2(\ln 3)} = \frac{4}{\ln 3}.
$$

- The variance

$$
\sigma^2 = \mu'_2 - \mu^2 = \frac{4}{\ln 3} - \frac{4}{(\ln 3)^2} = \frac{4\,(\ln 3 - 1)}{(\ln 3)^2}.
$$

Exercise 6.
A device that continuously measures and records seismic activity is placed in a remote region. The time, T, to failure of this device is exponentially distributed with mean 3 years, that is

$$
f(t) = \begin{cases} \frac{1}{3}e^{-t/3} & 0 < t < \infty \\ 0 & \text{elsewhere} \end{cases}
$$

Since the device will not be monitored during its first two years of service, the time to discovery of its failure is $X = \max(T, 2)$.

Calculate $E(X)$

Solution 6.

$$\begin{aligned} E[X] = E[\max(T,2)] &= \int_0^2 \frac{2}{3}e^{-t/3}\,dt + \int_2^\infty \frac{t}{3}e^{-t/3}\,dt \\ &= -2e^{-t/3}\Big|_0^2 - te^{-t/3}\Big|_2^\infty + \int_2^\infty e^{-t/3}\,dt \\ &= -2e^{-2/3} + 2e^{-2/3} - -3e^{-t/3}\Big|_2^\infty \\ &= 2 + 3e^{-2/3} \end{aligned}$$

Exercise 7.

Claim amounts for wind damage to insured homes are mutually independent random variables with common density function

$$f(x) = \begin{cases} \frac{3}{x^4} & x > 1 \\ 0 & \text{elsewhere} \end{cases}$$

where x is the amount of a claim in thousands.

Calculate the expected value of the largest of the three claims..

Solution 7.

To calculate the expected value of the largest of three claims, we denote the claims as X_1, X_2, X_3, which are independent and identically distributed (i.i.d.) random variables with the density function $f(x)$.

First, we find the cumulative distribution function (CDF) $F(x)$:

$$\begin{aligned} F(x) = P(X \le x) &= \int_1^x f(t)\,dt = \int_1^x \frac{3}{t^4}\,dt \\ &= \left[-\frac{1}{t^3}\right]_1^x = -\frac{1}{x^3} - (-1) = 1 - \frac{1}{x^3} \quad \text{for } x > 1 \end{aligned}$$

Thus, the CDF is:

$$F(x) = \begin{cases} 0 & x \le 1 \\ 1 - \frac{1}{x^3} & x > 1 \end{cases}$$

Second, find the CDF of the maximum $Y = \max(X_1, X_2, X_3)$

$$\begin{aligned} G_Y(y) = P(Y \le y) &= P(X_1 \le y, X_2 \le y, X_3 \le y) \\ &= P(X_1 \le y)P(X_2 \le y)P(X_3 \le y) \\ &= F(y)^3 \\ &= \left(1 - \frac{1}{y^3}\right)^3 \quad \text{for } y > 1 \end{aligned}$$

Find the PDF of the maximum Y

$$g_Y(y) = \frac{d}{dy}G_Y(y) = \frac{d}{dy}\left(1 - \frac{1}{y^3}\right)^3$$
$$= 9\left(1 - \frac{1}{y^3}\right)^2 \cdot \frac{1}{y^4}$$

Thus, the PDF of Y is:

$$g_Y(y) = \begin{cases} 0 & x \le 1 \\ \frac{9\left(1-\frac{1}{y^3}\right)^2}{y^4} & y > 1 \end{cases}$$

Calculate the expected value $E[Y]$
The expected value of the maximum is given by:

$$E[Y] = \int_1^\infty y \cdot \frac{9\left(1 - \frac{1}{y^3}\right)^2}{y^4}\, dy$$
$$= 9\int_1^\infty \frac{\left(1 - \frac{1}{y^3}\right)^2}{y^3}\, dy$$
$$= 9\int_1^\infty \left(\frac{1}{y^3} - \frac{2}{y^6} + \frac{1}{y^9}\right) dy$$
$$= 9\left(\frac{1}{2} - \frac{2}{5} + \frac{1}{8}\right)$$
$$= \boxed{\frac{81}{40}}.$$

Exercise 8.
Find the moment-generating function of the discrete random variable X that has the probability distribution

$$f(x) = 2\left(\frac{1}{3}\right)^x, \qquad \text{for } x = 1, 2, 3, \ldots$$

and use it to determine the values of μ_1' and μ_2'.

Solution 8.

- The mgf can be derived as

$$M_X(t) = \sum_x e^{tx} 2\left(\frac{1}{3}\right)^x = 2\sum_{x=1}^{\infty}\left(\frac{e^t}{3}\right)^x = 2\frac{e^t/3}{1 - e^t/3} = \frac{2e^t}{3 - e^t}$$

- The first and second derivatives of $M_X(t)$ are

$$M_X'(t) = \frac{(3 - e^t)2e^t - 2e^t(-e^t)}{(3 - e^t)^2} = \frac{6e^t}{(3 - e^t)^2}$$

$$M_X''(t) = \frac{(3 - e^t)^2 6e^t - 6e^t \cdot 2(3 - e^t)(-e^t)}{(3 - e^t)^4}$$

- Thus,

$$\mu_1' = M_X'(0) = \frac{3}{2} \quad \text{ and } \mu_2' = M_X''(0) = \frac{24 - 12 \cdot 2(-1)}{16} = 3$$

- Note that the variance can be easily obtained as follow

$$\sigma^2 = \mu_2' - (\mu_1')^2 = 3 - \frac{9}{4} = \frac{3}{4}$$

Exercise 9.
Let X be a random variable with the following pdf

$$f(x) = \begin{cases} e^{-2x} + \frac{1}{2}e^{-x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

1. Calculate the moment generating function for X.
2. Use it to compute the first, second moments and variance of X.

Solution 9.

1. The moment generating function (MGF) is defined as

$$M_X(t) = E[e^{tX}] = \int_0^\infty e^{tx} f(x)\, dx.$$

Substitute $f(x)$:

$$M_X(t) = \int_0^\infty e^{tx} \left(e^{-2x} + \frac{1}{2}e^{-x} \right) dx = \int_0^\infty \left(e^{-(2-t)x} + \frac{1}{2}e^{-(1-t)x} \right) dx.$$

For convergence, we require $t < 1$. Then,

$$\int_0^\infty e^{-ax}\, dx = \frac{1}{a}, \quad a > 0.$$

Hence,

$$M_X(t) = \frac{1}{2-t} + \frac{1}{2(1-t)}, \quad t < 1.$$

$$\boxed{M_X(t) = \frac{1}{2-t} + \frac{1}{2(1-t)}, \quad t < 1.}$$

2. The moments are given by

$$E[X] = M_X'(0), \qquad E[X^2] = M_X''(0).$$

First derivative:

$$M_X'(t) = \frac{1}{(2-t)^2} + \frac{1}{2(1-t)^2}.$$

Then

$$E[X] = M_X'(0) = \frac{1}{2^2} + \frac{1}{2(1)^2} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}.$$

$$\boxed{E[X] = \frac{3}{4}.}$$

Second derivative:

$$M_X''(t) = \frac{2}{(2-t)^3} + \frac{1}{(1-t)^3}.$$

Then

$$E[X^2] = M_X''(0) = \frac{2}{2^3} + \frac{1}{1^3} = \frac{1}{4} + 1 = \frac{5}{4}.$$

$$\boxed{E[X^2] = \frac{5}{4}.}$$

Variance

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = \frac{5}{4} - \left(\frac{3}{4}\right)^2 = \frac{11}{16}.$$

$$\boxed{\mathrm{Var}(X) = \frac{11}{16}.}$$